

Production Monitoring & Alerting System

You are hired as a **DevOps Engineer** for a startup running an online service (e.g., a food delivery / e-commerce / SaaS platform).

Your responsibility is to design and deploy a **monitoring and alerting system** that ensures:

- System availability
- Application performance
- External endpoint uptime
- Immediate alerting to the operations team via Slack

Objective

Build a **complete monitoring pipeline** using:

- Prometheus
- Grafana
- Alertmanager
- Blackbox Exporter
- Node Exporter
- A sample application (Python or any service)

System Architecture (what students must build)

App / Server / Website

↓

Exporters (Node + Blackbox + App)

↓

Prometheus

↓

Alertmanager

↓

Slack

↓

Grafana Dashboard

Tasks

Task 1: Setup Monitoring Stack

- Deploy Prometheus, Grafana, Alertmanager using Docker Compose

- Ensure all services are connected via a Docker network

✓ Deliverable:

- docker-compose.yml

Task 2: System Monitoring (Node Exporter)

- Monitor CPU, memory, disk usage
- Add Prometheus scrape config

✓ Queries to demonstrate:

```
rate(node_cpu_seconds_total[1m])
node_memory_MemAvailable_bytes
```

✓ Deliverable:

- Screenshots of metrics in Prometheus and Grafana

Task 3: Application Monitoring

- Create a small app (Python preferred) exposing:
 - Request count
 - Latency (Histogram)
 - Custom metric (e.g., temperature / queue size)

✓ Deliverable:

- Code snippet + /metrics endpoint output

Task 4: External Monitoring (Blackbox Exporter)

- Monitor at least **2 endpoints**, for example:
 - <https://www.google.com>
 - Your application endpoint

✓ Metrics to use:

```
probe_success
probe_duration_seconds
```

✓ Deliverable:

- Prometheus targets page screenshot
- Query results

Task 5: Alert Rules (Prometheus)

Create at least **3 alert rules**:

Example alerts:

1. High CPU usage

```
100 * (1 - avg(rate(node_cpu_seconds_total{mode="idle"}[5m])) by (instance)) > 80
```

2. Endpoint down

```
probe_success == 0
```

3. High latency

```
histogram_quantile(0.95, rate(app_latency_seconds_bucket[5m])) > 1
```

✓ Deliverable:

- alert.rules.yml

Task 6: Alertmanager + Slack Integration

- Configure Slack webhook in Alertmanager
- Alerts must be sent to a Slack channel

✓ Deliverable:

- Screenshot of Slack alert message

Task 7: Grafana Dashboard

Create a dashboard with:

- CPU usage per instance
- Memory usage
- Request rate
- Latency (p95)
- Endpoint availability

✓ Deliverable:

- Dashboard screenshot
- JSON export of dashboard

Evaluation Criteria

Criteria	Marks
Setup & architecture	20
Metrics collection	20
PromQL usage	20
Alerting correctness	20
Grafana dashboards	10
Documentation	10

Submission Requirements

Students must submit:

1. GitHub repository (code + configs)
2. Screenshots:
 - Prometheus targets
 - Grafana dashboards
 - Slack alerts
3. README explaining:
 - Architecture
 - PromQL queries
 - Alerts logic